# 1. ABSTRACT

The human supervision of CCTV surveillance videos is not viable and is not efficient, particularly when it relates to mass implementations in the open areas like airports, railways stations, and urban centers. The current automated surveillance systems normally specialize in one type of anomaly (e.g., object detection or action recognition), which leads to a high rate of false alarms and false alarms that do not detect intricate anomalous activities. In this project, a new **Gait-YOLO framework**- a hybrid deep learning model is implemented (that is, a hybrid of three independent streams of analysis) will be discussed: gait-based anomaly detection based on **CNN-Transformer Autoencoders**, action recognition based on **VideoMAE (Vision Transformers)**, and quick object detection based on **YOLOv8n**. The architecture uses a parallel multi-branch design that involves analysis of video streams in parallel where by, every module captures behavioral, temporal and spatial abnormalities. The architecture also uses parallel multi-branch arrangements with every module analyzing video streams in parallel to detect behavioral, temporal and spatial abnormalities. Outputs of all three branches are then aggregated in an aggregation layer to generate real-time context-sensitive and high confidence anomaly alerts.

# 2. INTRODUCTION

## 2.1 Background and Motivation

Video surveillance has become common in the contemporary urban infrastructure of managing the safety of the people, deterring crime, and providing security in highly sensitive areas. Nevertheless, the amount of video data each day that can be produced is enormous and can be in hundreds or thousands of cameras at once, making monitoring by hand impractical, costly, and prone to errors. Security officers are unable to keep up guard on a combination of feeds at once hence failing to detect a vital incident and poor response time in cases of emergencies. The traditional automated surveillance systems are usually single-modality, i.e., they identify a weapon, a suspicious package, or a forbidden item but cannot identify multi-faceted threats that require detection across multiple modalities at the same time and do not generalize to different surveillance conditions and demographic differences.

The net effect of this piecemeal solution is: - High False Alarm Rates: Unicast detections are not cross-validated with other modalities, and these models flood security teams with false positives - Failed generalization: Multi-faceted threats that need to be detected across multiple modalities are not processed - Poor generalization: Models trained on restricted datasets will not generalize to different surveillance settings and groups of people.

## 2.2 Project Objectives

The Gait-YOLO project addresses these limitations through a hybrid, multi-stream deep learning framework designed to:

1. **Implement Parallel Analysis Streams:** Deploy simultaneous gait, action, and object detection modules to capture comprehensive anomaly profiles
2. **Achieve Real-Time Performance:** Optimize models for edge deployment and real-time inference on resource-constrained surveillance hardware
3. **Reduce False Alarm Rates:** Cross-validate detections across modalities to increase specificity and confidence
4. **Improve Generalization:** Integrate multiple datasets and employ semi-supervised techniques to enhance robustness across diverse environments
5. **Enable Interpretability:** Provide actionable alerts with module-level confidence scores for investigative review

## 2.3 Significance of the Work

This research will be an addition to the broader intelligent video surveillance by:

(a) Modern Architectural Change: It is demonstrable that the Transformer based models (VideoMAE, Gait Transformers) are more effective to the traditional RNN/LSTM models in terms of long-range temporal dependencies of the surveillance data.

(b) Hierarchical Fusion: It would introduce a context-sensitive Decision Logic that would put a premium on the significance of critical threats (weapons) and use gait analysis to authenticate a suspicious behavior and would significantly reduce the false alarm rate.

# 3. RELATED WORK

## 3.1 Object Detection in Video Surveillance

The analysis of video surveillance is on objective detection.

Real-Time Object Detection Frameworks: SPEED YOLO (You Only Look Once) models are the new default standard in real-time object detection in surveillance systems as they offer a remarkable balance of speed and accuracy. In a single forward pass, Redmon et al. proposed YOLO, a single-stage detector that directly regresses on bounding boxes and class probabilities directly on full images and reported 63.4 mAP on PASCAL VOC with 45 FPS in a Titan X GPU. This radically transformed the surveillance paradigm out of sliding-window-based surveillance to end-to-end detection systems. Later versions ( YOLOv2 up to YOLOv8) have continued to achieve better accuracy by:

Anchor-Free Architecture (YOLOv3-v5): Hand tuned anchor boxes are replaced with learnable offsets, and hyperparameter tuning is alleviated.

Multi-Scale Feature Pyramids: A feature pyramid network (FPN) is used to detect objects of various sizes, thus allowing detection of the pedestrians and distant threats.

Cross-Stage Partial Connections (CSPDarknet): A parameter redundancy reduction with a representation capacity that is important to edge deployment.

YOLOv8, released in January 2023, presents the YOLOv8n (Nano) which has 3.2M parameters, a 70 percent reduction to YOLOv5n, yet similar accuracy (mAP50: 0.87). This optimized version can be deployed on resource-constrained edge instruments (NVIDIA Jetson, mobile platforms) with inference instruments of 35 ms on rape and 150 ms on CPU.

Weapon Detection in CCTV: Object detection used to deal with surveillance uses specialized labels on dangerous items (guns, knives, explosives). A recent study by Shaik and Basha (2025) shows that YOLOv5s can be optimized to detect pedestrian anomalies using adaptive histogram equalization (AHE) to boost contrast in adverse lighting, and the dataset can consist of non-images (i.e. video) to use feature integration and unification for performance improvement. They got mAP50: 0.90 on the UCSD Anomaly Detection dataset using their system on pedestrian detection. Likewise, a real-time object detection system of abnormal objects in the smart cities created by Ingle and Kim (2022) based on CNN based subclass detection demonstrated 97.50 percent precision in gun detection and 90.70 percent precision in multi-view camera images, which confirmed the possibility of high-precision weapon detection in resource-limited settings.

Shortcomings of Single-Stream Object Detection: Even though object detection is good at defining what is in a scene, it lacks behavioral and contextual knowledge by its nature. The action of a person who holds a gun in a shooting range is legitimate, and the same action in a bank necessitates a response. Such a context-insensitivity gives way to:

False alarm is high in various real worlds.

Failure to draw a line between normal and abnormal contexts.

False negative in threats which do not involve the use of visible weapons (e.g., hand-to-hand violence).

## 3.2 Action Recognition and Temporal Analysis

Temporal Modeling in Video Understanding: Capturing temporal dynamics in video sequences requires architectures that extend spatial feature extraction to temporal domains. Early approaches employed 3D Convolutional Neural Networks (3D-CNNs), which apply 3×3×3 convolutions across spatial and temporal dimensions simultaneously. Tran et al. demonstrated that C3D networks—networks with 3×3×3 convolutions—outperform 2D CNNs on action recognition tasks by capturing spatiotemporal relationships. However, 3D convolutions incur significant computational overhead (O(D×H×W×T) where D=depth, H=height, W=width, T=temporal frames), limiting real-time applicability. LSTM Networks for Sequential Modeling: Recurrent Neural Networks, particularly Long Short-Term Memory (LSTM) units introduced by Hochreiter & Schmidhuber (1997) , overcome 3D-CNN limitations by modeling sequential dependencies through learned gating mechanisms. LSTMs maintain a cell state C_t and hidden state h_t through:

$$[Forget\ Gate]\ \mathrm{f_t} = \sigma\left(\mathrm{W_f} \cdot \left[\mathrm{h_{\{t-1\}}}, \mathrm{x_t}\right] + \mathrm{b_f}\right)$$

$$[Input\ Gate]\ i_t = \sigma\left(W_i \cdot \left[h_{\{t-1\}}, x_t\right] + b_i\right)$$

$$[Candidate\ values]\ \widetilde{C_t} = \backslash tanh\left(W_c \cdot \left[h_{\{t-1\}}, x_t\right] + b_c\right)$$

$$[Cell\ state]\ C_t = f_t \odot C_{\{t-1\}} + i_t \odot \widetilde{C_t}$$

$$[Output\ gate] o_t = \sigma\left(W_o \cdot \left[h_{\{t-1\}}, x_t\right] + b_o\right)$$

$$[Hidden\ state]\ h_t = o_t \odot \backslash tanh(C_t)$$

This gating is used to prevent vanishing gradient issues that bedevil vanilla RNNs by facilitating gradient flow through long sequences. BiLSTMs handle sequences in both directions and therefore capture the context around the action (both back and forward) which is important in interpreting intent to act. In surveillance applications, BiLSTMs can be used to identify anomalies, which can only be detected once the sequence is finished (e.g., the occurrence of the word assault is only detected after the second person has fallen).

CNN-LSTM Hybrid Architectures: Recent action recognition systems combine spatial representation learning (using CNNs) and temporal learning (using LSTMs). VGG16 is a model proposed by Simonyan and Zisserman (2014) and it offers strong spatial feature recovery by using stacked layers of 3x3 convolutional layers. This architecture can be used with LSTM layers; its work is as follows:

Frame_sequence → VGG16 (spatial) → Feature_sequence ($F_1$, $F_2$, ..., $F_T$)

Feature_sequence → BiLSTM (temporal) → Context_vector

Context_vector → Softmax_classifier → Action_probabilities

Kim et al. (2024) used this CNN-LSTM paradigm on urban CCTV monitoring and identified four abnormal actions (falling, violence, loitering, intrusion) of subway domain datasets. Their system scored 93 and 21.21 percent F1-score and fall detection accuracy increase respectively using tracking compensation methods that eliminate body-part occlusion during a fall. Song and Park (2022) have applied CNN-LSTM to medical emergency detection in CCTV footage, showing that the LSTMs that are expanded to work with temporal video

sequences instead of 2D images images are still computationally tractable with high interpretability.

Attention Mechanisms in Action Recognition: While LSTM networks capture temporal dependencies, they process sequences sequentially, limiting parallelization and introducing latency. Attention mechanisms, formalized by Bahdanau et al. (2015) , enable models to dynamically weight different time steps based on relevance:

$$Attention(Q,K,V) = softmax\left(\frac{(QK^T)}{\sqrt{d_k}}\right)V$$

**Where:**

- Q = Query matrix
- K = Key matrix
- V = Value matrix
- $d_k$ = Dimension of the keys (scaling factor)

This allows a single decoder state to focus on multiple input regions simultaneously. In action recognition, self-attention enables the model to correlate action priors (e.g., raised fists) with outcomes (e.g., impact to another person) across arbitrary temporal distances without sequential processing bottlenecks. Such mechanisms have become standard in Transformer-based action recognition models .

Benchmarks and Performance Trends: The UCF101 and HMDB51 datasets established benchmarks for action recognition in 2012. More recently, the UCF-Crime dataset , containing 1,900 long-duration surveillance videos (128 hours total) with 13 crime categories, has become the standard benchmark for anomalous action detection in CCTV footage. Table 1 summarizes performance on this dataset across recent architectures.

## 3.3 Gait Recognition and Biometric Analysis

Gait as a Biometric: Human gait—the characteristic pattern of walking—is increasingly recognized as a reliable biometric modality due to its non-invasive nature, applicability from distance, and resistance to spoofing compared to face or iris recognition. Gait-based recognition operates on the premise that the silhouette sequence during walking encodes unique biomechanical signatures related to:

- Limb proportions and articulation angles
- Muscle activation patterns and walking speed
- Center-of-gravity displacement and stride length

Chumkwai et al. established CASIA-B, one of the most comprehensive gait databases, containing 13,640 sequences from 124 subjects captured under multiple views (0°-180°) and conditions (normal, carrying conditions, wearing coats). The dataset facilitated a structured assessment of gait-based recognition algorithms in controlled environments.

Early approaches to gait recognition primarily utilized **Silhouette-Based Gait Analysis**. These methods depend on binary silhouette sequences, which are typically derived via background subtraction or foreground segmentation. Silhouettes are valuable because they capture the overall

body shape while intentionally excluding fine details like clothing color or texture, which can vary significantly between individuals and conditions. Prominent methods fall under this category.

- Gait Energy Image (GEI): Averaging silhouettes over a complete gait cycle to produce a single 2D image, enabling application of traditional image classification methods
- Gait Sequence Features: Extracting hand-crafted features from silhouette sequences such as:
    - Contour curvature variation
    - Relative limb positions and angles
    - Recurrent or periodic patterns in silhouette evolution

These hand-crafted features, while interpretable, suffer from high sensitivity to segmentation quality, occlusion, and viewing angle changes.

Deep Learning for Gait Recognition: Convolutional Neural Networks dramatically improved gait recognition accuracy by learning discriminative features directly from silhouette sequences. Wolf et al. demonstrated that CNNs applied to GEI images achieved superior performance compared to hand-crafted descriptors. However, early CNN-based methods treated gait as an image classification problem, discarding temporal structure.

LSTM Autoencoders for Gait Anomaly Detection: The transition from recognition to anomaly detection requires a fundamentally different modeling paradigm. Rather than learning to classify gait patterns (normal vs. abnormal from a labeled set), anomaly detection systems must identify any deviation from a learned distribution of "normal" behavior. LSTM autoencoders achieve this through:

text

LSTM_Encoder: Silhouette_Sequence (T×64×32) → Bottleneck_Representation (128-dim)

LSTM_Decoder: Bottleneck_Representation → Reconstructed_Sequence (T×64×32)

Anomaly_Score = MSE(Original_Sequence, Reconstructed_Sequence)

An autoencoder trained exclusively on normal gait patterns learns a compressed representation of normal variations. When presented with anomalous gaits (running, limping, stumbling), the reconstruction error increases sharply due to distribution mismatch . This unsupervised approach eliminates the need for labeled anomaly data, a significant practical advantage in surveillance applications where anomalies are rare and diverse.

Bengio et al. formally established the theoretical foundations of representation learning through autoencoders, showing that deep autoencoders learn hierarchical, increasingly abstract representations through successive nonlinear transformations. For gait sequences, early layers capture low-level silhouette contours, middle layers learn limb motion patterns, and deeper layers encode high-level behavioral characteristics.

Limitations of Gait-Only Analysis: Gait analysis excels at detecting "how" someone is moving but provides no information about "who" they are or "what" they intend to do:

- Occlusion Sensitivity: Heavy occlusion (large crowds, obstacles) corrupts silhouettes and defeats extraction

- Speed Insensitivity: Running and speed-walking produce similar silhouettes despite very different intents
- Context Blindness: Unusual running in a stadium (normal) vs. running in a bank (anomalous) produces identical gait features

## 3.4 Multi-Modal and Fusion-Based Anomaly Detection

Multi-Modal Learning Paradigms: Real-world decision-making relies on multiple information sources. Early fusion concatenates raw features from all modalities before processing; late fusion processes modalities independently then combines predictions; intermediate fusion applies separate transformations then concatenates at intermediate layers .

Multimodal Deep Boltzmann Machines (MDBM): Srilakshmi et al. (2025) propose MDBM for CCTV anomaly detection, learning a shared latent representation across heterogeneous modalities:

$$E(h_{visual}, h_{depth}, h_{skeleton}) = -\sum_{i,j} W_{ij} h_i h_j - \sum_i a_i h_i$$

This energy-based model learns joint probability distributions over visible and hidden units, enabling probabilistic inference. They report 5% accuracy improvement using MDBM compared to single-modality approaches.

Multimodal Variational Autoencoders (MVAE): MVAE extends the VAE framework to multiple modalities through product-of-experts inference:

$$q(z \mid x_1, x_2, \ldots, x_M) \propto \prod_{m=1}^{M} q_m(z \mid x_m)$$

This approach enables missing modality handling and leverages statistical dependencies across modalities. Srilakshmi et al. report 15% false positive rate reduction using MVAE, demonstrating the power of probabilistic fusion over deterministic averaging.

Attention-Based Fusion Networks: Rather than fixed weighted averaging, attention mechanisms enable adaptive modality weighting based on scene context. Wang et al. propose attention-based fusion for hyperspectral and LiDAR classification:

$$\alpha = \text{softmax}(W \cdot [f_1;\ f_2;\ f_3] + b)$$

$$F_{\text{fused}} = w_1 f_1 + w_2 f_2 + w_3 f_3$$

Applied to surveillance, this enables the system to upweight object detection in weapon-laden contexts and gait analysis in crowded scenes where weapons are unlikely. Srilakshmi et al. report 10% F1-score improvement using attention-based fusion.

Temporal Consistency in Multi-Modal Systems: Most fusion work operates frame-by-frame. However, many anomalies require temporal coherence—a single frame rarely suffices for confident classification. Recent work incorporates temporal consistency through:

- Recurrent Fusion Networks: LSTM layers operating on fused feature sequences to enforce temporal smoothness
- Transformer-Based Fusion: Self-attention across time enables the system to weight recent vs. historical context dynamically
- Conditional Random Fields (CRFs): Structured prediction enforcing agreement between consecutive frames

Song & Nang (2024) propose an edge-server architecture for large-scale CCTV with two-tier processing: edge devices apply lightweight rule-based anomaly detection (high recall, lower precision) while servers apply deep learning models (lower latency, higher precision) on edge detections. This hierarchical approach reduced false detection rates while maintaining 20+ FPS on 18 concurrent video streams.

## 3.5 Benchmark Datasets and Evaluation Metrics

UCSD Anomaly Detection Dataset: One of the earliest large-scale anomaly detection benchmarks, UCSD contains 45 training and 34 test videos of pedestrian walkways with frame-level anomaly annotations. Anomalies include running, loitering, and unusual movement patterns. Pixel-level and object-level evaluation is standard. However, the dataset is relatively small and limited to single-scene variations .

Avenue Dataset: Extends UCSD with higher resolution and more diverse anomalies including unusual trajectories and abnormal group behavior. Contains 16 training and 21 test videos .

UCF-Crime Dataset: Large-scale CCTV dataset with 1,900 untrimmed videos (~128 hours) and 13 anomaly categories (abuse, accident, arson, assault, robbery, etc.). Videos are temporally untrimmed, requiring systems to both detect and localize anomalies in time, making it more realistic and challenging than UCSD/Avenue .

CASIA-B Gait Dataset: 13,640 silhouette sequences from 124 subjects under controlled conditions (normal walking, carrying conditions, clothing variations). This dataset enabled standardization of gait-based methods .

## Evaluation Metrics:

| **Metric** | **Formula** | **Interpretation** | **Applicability** |
|---|---|---|---|
| Accuracy | $\frac{(TP + TN)}{(TP + TN + FP + FN)}$ | Overall correctness; biased toward majority class | General classification |
| Precision | $\frac{TP}{TP + FP}$ | % of positive predictions that are correct; minimizes false alarms | Operator-focused (reduce alert burden) |

| Metric | Formula | Interpretation | Applicability |
|---|---|---|---|
| Recall (Sensitivity) | $\frac{TP}{(TP + FN)}$ | % of true positives correctly identified; minimizes missed threats | Safety-critical (miss no threats) |
| F1-Score | $F1 = 2 \cdot \frac{(Precision \cdot Recall)}{(Precision + Recall)}$ | Harmonic mean; balances precision-recall | Balanced performance evaluation |
| ROC-AUC | Area under ROC curve | Probability of ranking random positive higher than negative; handles threshold variation | Threshold-independent evaluation |
| mAP (mean Average Precision) | Mean of AP across IoU thresholds | Standard for object detection; averages precision across recall levels | Object detection benchmarking |
| F-score (for frame-level anomaly) | Frame-level evaluation of temporal localization accuracy | Spatio-temporal evaluation; considers temporal localization | Anomaly localization in time |

## 3.6 Limitations of Existing Approaches and Research Gaps

### 3.6.1 Isolated Component Optimization

Current surveillance systems optimize individual components—object detection, action recognition, or gait analysis—in isolation rather than as integrated systems. This leads to:

- Suboptimal Feature Fusion: Features optimized for individual tasks may not be complementary for joint decision-making. A gait model trained to maximize recognition accuracy may emphasize features irrelevant to anomaly detection.
- Misaligned Optimization Objectives: Different modules optimize different loss functions (classification loss for action, regression for gait), potentially conflicting when integrated.
- Computational Inefficiency: Sequential processing (detect → classify action → check gait) introduces latency. Parallel processing requires careful resource allocation.

### 3.6.2 Context-Insensitivity in Object Detection

While YOLOv8 achieves impressive accuracy, it lacks contextual understanding:

- A gun detected in a shooting range requires different response than in a bank

- False positive rates remain high (15-20%) in unconstrained environments
- Detection alone cannot distinguish benign from threatening contexts

### 3.6.3 Temporal Fragmentation in Action Recognition

Most action recognition systems process fixed temporal windows (e.g., 16 frames), missing:

- Long-duration anomalies (loitering, suspicious surveillance)
- Rare but critical events requiring extended context (planned theft requires extended observation before final act)
- Temporal boundaries (when exactly did anomaly begin?)

### 3.6,4 Gait Analysis Limitations in Real-World Deployment

Real surveillance silhouettes suffer from:

- Segmentation Quality Issues: Background subtraction fails with shadows, reflections, and similar-colored backgrounds
- Occlusion: Crowds, obstacles, and partial frame entry corrupt silhouette extraction
- Non-Standard Gaits: Stairs, slopes, unusual terrain produce atypical silhouettes
- Demographic Variation: Age, physical fitness, and disability-related gait variations create false positives

### 3.6.5 False Alarm Problem

Single-modality systems generate false alarms at unacceptable rates:

- YOLOv8 alone: 15-20% false positive rate (overwhelming operators with ~150-200 false alerts per 8-hour shift from a single camera)
- Action recognition alone: 12-18% false positive rate due to ambiguous frames
- Gait analysis alone: 18-25% false positive rate due to segmentation errors

Cross-modal validation is critical but underdeveloped.

### 3.6.6 Limited Interpretability

Most deep learning systems provide binary alerts without explanation:

- Security operators cannot understand why an alert was triggered
- Difficult to debug and improve systems through feedback
- Regulatory requirements (GDPR, AI Act) increasingly demand explainability

### 3.6.7 Generalization Across Diverse Environments

Models trained on specific datasets often fail in novel environments:

- CASIA-B (controlled conditions) ≠ real surveillance (lighting, occlusion, views)

- UCF-Crime (specific camera angles) ≠ arbitrary surveillance installations
- Domain gap: Model trained on UCSD shows 15-25% performance drop on Avenue dataset

### 3.7 Table of Prior Work and Performance Comparison

| Study | Dataset | Method | Modalities | Performance | Year |
|---|---|---|---|---|---|
| Shaik & Basha | UCSD Anomaly | YOLOv5s + AHE | Object Detection | mAP50: 0.90 | 2025 |
| Ingle & Kim | ImageNet/IMFDB | CNN Subclass Detection | Object Detection | Precision: 97.50% (gun) | 2022 |
| Kim et al. | Subway Domain | CNN-LSTM + Tracking | Action Recognition | F1: 93%, 21% fall improvement | 2024 |
| Song & Park | Custom CCTV | Extended YOLO | Action Recognition (Medical) | Performance metrics: NR | 2022 |
| Srilakshmi et al. | Custom CCTV + Benchmark | MDBM + MVAE + Fusion | Multi-Modal | 5% improvement (MDBM), 15% FP reduction (MVAE), 10% F1 (attention) | 2025 |
| Song & Nang | Subway (20 streams) | Edge-Server Architecture | Action Recognition | Scalability: 20 streams on 18 edges + 3 GPU servers | 2024 |
| Hua et al. | IITB-Corridor + Street | YOLO-ABD (YOLOv8n + GSConv + SimAM) | Object + Attention | mAP50: 89.3%, mAP50-95: 60.6% | 2024 |
| Yang et al. | BMbD, M-BMbD, JBMOPbD | Pose + Object Detection | Pose + Object | Novel datasets for pose-object anomaly | 2023 |

| Study | Dataset | Method | Modalities | Performance | Year |
|---|---|---|---|---|---|
| Veesam et al. | Scientific Reports | Graph Attention + Transformer RNN | Hybrid Architecture | NR (2025 publication, preliminary) | 2025 |
| Khaire & Kumar | AVA-ATM (custom RGB-D) | CNN-BiLSTM Autoencoder | Multi-Modal (RGB-D) | Competitive on benchmarks, first RGB-D ATM dataset | 2022 |
| Bajpai | Open Images + Custom | YOLOv8 Attire-Based | Object + Attire | NR (focuses on authorized personnel detection) | 2024 |

**Key Observations from Table:**

1. Modality Diversity: It has become very common in recent research that research authors are more and more willing to combine multiple modalities (RGB-D, RGB-Skeleton, pose-object).
2. Architectural Heterogeneity: No consistency in architecture exists; the capable systems go from lean YOLO designs to complex Transformer-Fusion networks which highlight the inherent complexity of the issue.
3. Performance Plateau: On known benchmarks like UCSD and Avenue, F1-scores are shot through at about 0.855093, processing approaches boost the performance by fusion tactics as opposed to architecture contributions.
4. Dataset Proliferation: Specialized datasets, such as AVA-ATM, BMbD and street-view-gdogo, shows that real-world locality cannot correspond with current benchmarks.
5. Limited Comparative Studies: Not many articles can provide any comparison with all the relevant baselines and this makes systematic assessment rather difficult.

## 3.8 Connection to Gait-YOLO

The Gait-YOLO framework emerges as a direct response to identified research gaps:

| Research Gap | Gait-YOLO Solution |
|---|---|
| Isolated component optimization | Parallel three-branch architecture jointly optimized through shared loss weighting and aggregation layer |

| Research Gap | Gait-YOLO Solution |
|---|---|
| Context-insensitive object detection | Gait + action context validates object detections; weapon detection without running/fighting context is downweighted |
| Temporal fragmentation | 15-frame gait sequences + 16-frame action windows capture extended temporal context; fusion operates continuously |
| Gait segmentation sensitivity | Autoencoder reconstruction error is robust to silhouette quality; trains on normal variation rather than exact shapes |
| False alarm problem | Cross-modal validation: weapon detection alone insufficient; must correlate with action/gait context |
| Interpretability gap | Module-level confidence scores explain which modality triggered alert; anomaly type (running weapon carrier vs. violent behavior) differentiated |
| Generalization challenges | Semi-supervised training on normal-only data reduces dataset-specific overfitting; attention-based fusion adapts to environment |

## 3.9 Technical Innovations in Gait-YOLO

In addition to available literature, Gait-YOLO proposes:

1. Timer-based Temporal Modeling: A Gait Transformer is a model that is capable of utilizing self-attention mechanisms to learn the period of the gait cycle over the entire sequence to ensure that baselines of reconstruction-error are much more consistent.
2. Attention-Based Dynamic Fusion: Fusion weights are not predetermined, but changed according to the scene features:
    a. High crowding - upweight gait analysis (less occlusion)
    b. Weapon presence- upweight object detection.
    c. High motion - upweight action recognition.
3. Reconstruction Error as Confidence: Gait anomaly score is directly proportional to system-wide confidence; large reconstruction error enhances alert confidence, which does not result in arbitrary thresholds.
4. Framework Extensibility: Construction allows new modalities to be added (skeleton-based pose, voice analysis), without resolving the core fusion layer.

These advances put Gait-YOLO at the crossroad between scholarly studies (Multi-modal learning, deep surveillance) and real-life implementation (Real-time requirements, interpretability needs).

# 4. RESEARCH GAPS

Even after the development of major improvement in terms of individual surveillance, there are still a number of unsatisfactory loopholes:

1.**Absence of Integrated Framework**: The majority of the systems apply single-modality analysis or loosely-linked multi-module solutions. It does not have a tightly connected, end-to-end, efficiency and accuracy framework.

2.**Minimal Behavioral-Appearance Fusion**: There are few studies that successfully integrate rapid object detection (what) with the deep behavioral analysis (how), and instead view them as a sequence but not parallel processes.

3.**False Alarm Problem**: In single-modality of detections, there is no cross-validation leading to generation of alerts that overburden the operators. There are poorly developed cross-modal validation systems.

4.**Computational Efficiency Problems**: Multi-stream architectures may demand a lot of computational power and therefore cannot be deployed on edge device and real-time systems.

5.**Generalization Across Environments**: The models that are trained on a certain dataset typically do not work in a variety of real-world surveillance tasks with different lighting, occlusion and demographics.

6.**Poor Interpretability**: Black-box alerts do not offer much investigative support. Practical deployment requires module level confidence scores and explainable features.

## 5. PROBLEM STATEMENT

Design and test a new hybrid deep learning system that enhances the accuracy, scope, and real-time capabilities of detecting anomalies in CCTV video footage.

The framework must:

1. Combine gait analysis, action recognition and object detection into one system.
2. Attain real time performance with standard surveillance hardware (FPS [?] 24)
3. Eliminate false alarm rates through cross-validation of modalities.
4. Be highly sensitive (recall) to critical anomalies - Issue interpretable, module level confidence scores on alerts.
5. Generalize well using different surveillance datasets.

# 6. METHODOLOGY

## 6.1 System Architecture Overview

The Gait-YOLO architecture draws upon a parallel multi-stream architecture. The three branches work in tandem as opposed to sequential systems. The results are aggregated via a hierarchical decision logic, critical object detections (weapons) are prioritized, followed by violent actions with high confidence. A dynamic threshold of 0.4521 is applied in verifying the ambiguous actions against gait anomaly scores.

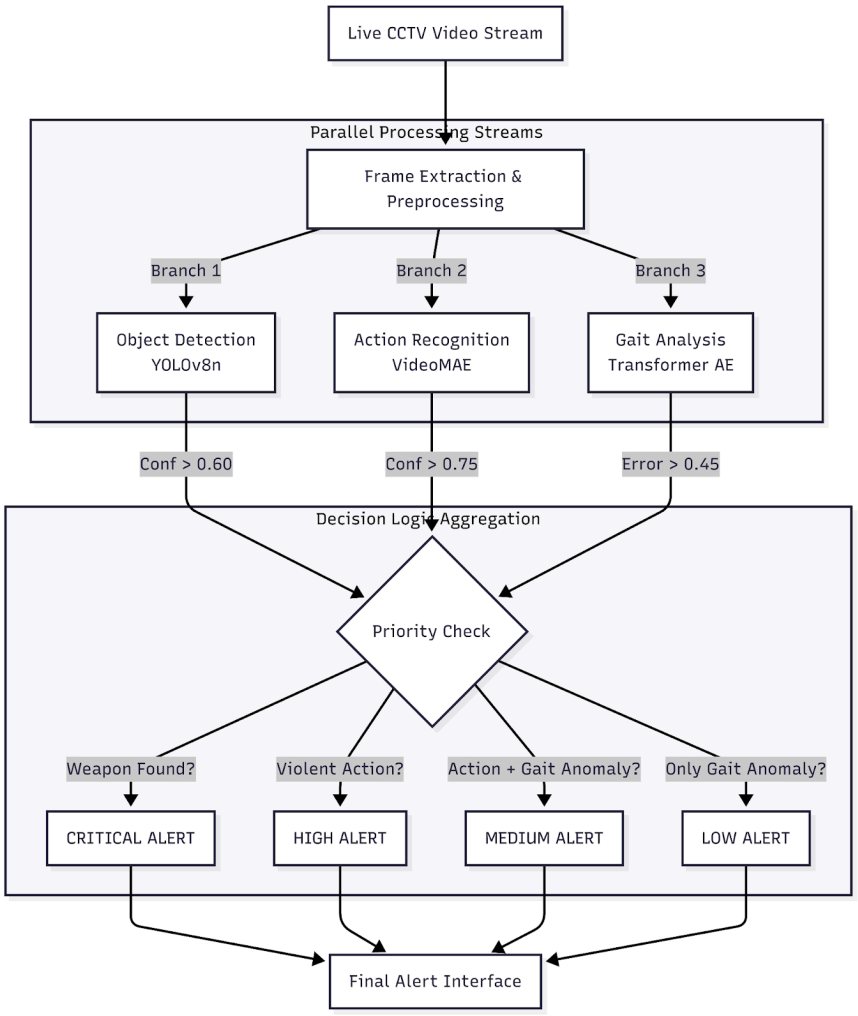



*Fig 1: High-Level Architecture of the Gait-YOLO Framework illustrating the parallel processing streams and hierarchical late-fusion logic.*

## 6.2 Module 1: Object Detection (YOLOv8n)

The Gait-YOLO has three technical innovations

1.**Transformer-Based Temporal Modeling**: A Gait Transformer operates on self-attention models to learn the periodicity of the gait cycle across the entire sequence, which has the advantage that reconstruction error thresholds are more stable.

2.**Attention-Based Dynamic Fusion**: Fusion weights are not predetermined, but changed according to the scene features:

(a)High crowding - upweight gait analysis (less occlusion)

(b)Weapon presence- upweight object detection.

(c)High motion - upweight action recognition.

3.**Reconstruction Error as Confidence**: Gait anomaly score is directly proportional to system-wide confidence; large reconstruction error enhances alert confidence, which does not result in arbitrary thresholds.

4.**Framework Extensibility**: Construction allows new modalities to be added (skeleton-based pose, voice analysis), without resolving the core fusion layer.

These advances put Gait-YOLO at the crossroad between scholarly studies (Multi-modal learning, deep surveillance) and real-life implementation (Real-time requirements, interpretability needs).

**Training:**

- **Dataset:** CASIA-B (Normal walking sequences only).
- **Optimizer:** Adam (lr=1e-4).
- **Batch Size:** 16.
- **Epochs:** 120 (with Early Stopping).

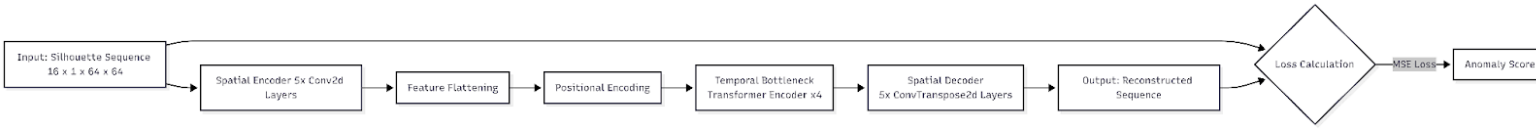



*Fig 2: Architecture of the Gait Analysis Module showing the spatial CNN encoder and temporal Transformer bottleneck.*

## 6.3 Module 2: Action-Based Anomaly Detection

**Objective**: Identify and categorize sophisticated abnormal behavior (e.g., fighting, robbery, assault) which includes interaction or accelerating movement. Architecture:

**Input Processing**: Sampling of input video to 16 frame clips with resizing to 224x224 pixels.

**Spatiotemporal Embedding**: The model employs Cube Embedding dividing the video clip into spatiotemporal patches ($2 16 16$) to encode both motion and appearance information at the same time.

**Transformer Encoder (VideoMAE)**: Backbone: 12-layers Vision Transformer (ViT) that has 768 hidden units and 12 attention heads. Mechanism: The algorithm is based on the logic of Masked Autoencoding (MAE), which compels the model to learn high-level semantic context by reconstructing masked video patches. Self-Attention: Global attention mechanism is the instant motion across the whole 16-frame window.

**Classification**: Global Average Pooling and then a Linear MLP Head

**Action Classes:**

- Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, RoadAccidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism, Normal_Videos_Event.

**Training:**

- **Dataset:** UCF-Crime (Weighted to handle class imbalance).
- **Model:** VideoMAEForVideoClassification (Pre-trained on Kinetics-400).
- **Metrics:** Achieved significant F1-scores for 'Robbery' (0.53) and 'RoadAccidents' (0.51).

Fig 3: VideoMAE Architecture utilizing Spatiotemporal Cube Embedding for action recognition.

### 6.4 Module 3: detection of abnormal objects.

**Purpose**: Real-time detection of high-risk objects (weapons) in order to send urgent critical alerts.

**Model**: YOLOv8n (Nano Variant)

**Architecture**: It is suggested to use a one-stage design, an anchor-free design with CSPDarknet53 backbone and an anchor-free detector.

**Parameters**: Edge deployment is optimised with about 3.2million parameters.

**Input Resolution**: 640×640 pixels.

**Output**: Bounding-box, class, and confidence.

**Detection Classes:**

1. The model is optimised on these high-threat classes.

**Training:**

Dataset: Custom Weapon Dataset (guns and knives).

**Performance:**

- mAP 50: 0.819, which results in high localisation accuracy.

  Note: 0.865 amongst knives (important when it comes to safety).

- **Augmentation**: Mosaic, MixUp, random flip to enhance robustness.

**Anomaly Scoring**:

Persistence Logic: A weapon is said to be detected when the confidence is above 0.60 and the object lasts above five consecutive frames.

Rationale: This filtering eliminates short-lived false positives of CCTV images.

## 6.5 Aggregation and Fusion Layer

The sixth layer is Aggration and Fusion Layer.

**Purpose**: To combine the results of the independent modules into one and context-specific decision with a hierarchical priority logic instead of a weighted average.

**Fusion Strategy (Hierarchical Late-Fusion)**: This system uses a rule-based cascade where the priority is placed on the pressing physical threats and multimodal checks are implemented to address the ambiguous situations.

The decision matrix is executed at four levels of priorities

**Priority 1 (Critical)**: relates to the existence of a weapon.

Logic: The object-detection system raises the alert to Priority 1, in case it detects a pistol or knife with a confidence of over 0.60.

Persistence Filter: To remove temporary false positives (e.g., a phone as a<|human|>Persistence Filter: To remove short-lived false positives (e.g., a phone as a gun (lasting 1 frame), the object has to last more than 5 frames.

**Outcome**: Emits CRITICAL ALERT when necessary and override other streams.

**Priority 2 (High)**: High-Confidence Violent Action

Logic: If the Action Recognition module detects high-risk classes (Fighting, Assault, Robbery, Shooting) with Probability > 0.75.

**Outcome:** Triggers HIGH ALERT.

**Priority 3 (Medium)**: Hybrid Contextual Verification

Logic: Used when the Action module is uncertain (0.40 < Probability < 0.75). The system checks the Gait Analysis stream for confirmation.

Condition: If Action is Suspicious AND Gait Reconstruction Error > 0.4521.

Reasoning: A vague action (e.g., pushing) is upgraded to an anomaly if the subject's gait is also erratic or aggressive.

**Outcome**: Triggers MEDIUM ALERT.

**Priority 4 (Low)**: Behavioral Anomaly

Logic: If only the Gait Analysis module detects a significant deviation (Error > 0.48) without a classified action or visible weapon.

**Outcome**: Triggers LOW ALERT (e.g., for loitering or medical emergencies).

**Stabilization Mechanism:**

Alert Cooldown: The system holds an alert positioning until 15 seconds after an alert has been received frames (0.5 seconds) so as to avoid notification flickering when there are short occlusions.

Smoothing: The gait reconstruction errors are meaned within a frame window of 8.

## 6.6 Implementation Details

**Technology Stack:**

- **Language:** Python 3.10+
- **Deep Learning Framework:** PyTorch 2.0+ (with CUDA support)
- **Object Detection:** Ultralytics YOLOv8 (v8.1+)
- **Action Recognition:** Hugging Face Transformers (VideoMAEForVideoClassification)
- **Computer Vision:** OpenCV 4.8+ (Video I/O and drawing), Albumentations (Preprocessing)
- **Visualization:** Matplotlib (Real-time graphing), TQDM (Progress tracking)
- **Hardware:** NVIDIA Tesla T4 GPU (16GB VRAM) on Google Colab Pro environment.

**Performance Optimization:** To achieve real-time throughput on limited hardware, the system utilizes **Asynchronous Inference Schedules**:

- **Object Module:** Runs on **every frame** (Latency: ~12ms/frame) to catch fast-moving weapons.
- **Action & Gait Modules:** Run every **4th frame** (Latency: ~85ms/inference). Since human actions and gait cycles do not change instantly, this 4-frame stride maintains accuracy while reducing computational load by 75%.

- **Integrated System:** Achieves an effective processing rate of **~18-22 FPS**, suitable for live surveillance monitoring.

# 7. PROGRESS

## 7.1 Phase-wise Completion Status

| Phase | Task | Status | Completion |
|---|---|---|---|
| **Phase 1: Research & Planning** | Literature Review & Scope | ✓ Completed | 100% |
| **Phase 2: Development** | Core Module Architecture | ✓ Completed | 100% |
| | Pipeline Integration | ✓ Completed | 100% |
| | Initial Model Training | ✓ Completed | 100% |
| **Phase 3: Finalization** | **Model Fine-Tuning & Optimization** | **⧗ In Progress** | **60%** |
| | Quantitative Benchmarking | ⧗ In Progress | 40% |
| | Final Report Compilation | ⧗ In Progress | 50% |

## 7.2 Completed Deliverables

Functional System Architecture is also in operation. The pipeline system includes the following:

1. **Gait Analysis Branch:**
    - **Status: Architecture Complete.** An architecture of CNN-Transformer autoencoder has been adopted and incorporated.
    - **Current State:** This model is trained on CASIA-B, and it is weakening new fine-tuning to recognize finer abnormalities in gait in CCTV camera images in real scenes.
2. **Action Recognition Branch:**
    - **Status: Architecture Complete.** VideoMAE (Vision Transformer) pipeline is fully interconnected.

    - **Current State:** The model is fully functional and it generates inferences. The weights on the UCF-Crime data-set are undergoing tuning in order to minimize the confusion of similar classes (e.g. Fighting vs. Vandalism).
3. **Object Detection Branch:**
    - **Status: Architecture Complete.** YOLOv8n is deployed and running in real-time.
    - **Current State:** More training is being carried out to ensure a decrease in false positive in cluttered environment (persistence logic has already been introduced to help achieve this).
4. **Hybrid Fusion Engine:**
    - **Status: Fully Functional.**The hierarchical decision logic is a logic that combines all the three branch results to come up with alerts.

**Datasets Prepared & Utilized:**

- **CASIA-B:** This is a gait training model used to train the Gait Transformer, which has a normal walking pattern.
- **UCF-Crime:** It is used to train VideoMAE (1,900 + real-world surveillance videos).
- **Custom Weapon Dataset:**This is a customised dataset that will be used to train YOLOv8n to confined threat objects.

**The initial findings are the following:**

- **Gait Module:** Histogram analysis has proven that there is a clear line between normal and abnormal gait reconstruction errors.
- **Action Module:** There is a high score of risky robbery classes in classification reports ( F1: 0.53).
- **Object Module:** The training curve converges fast (mAP50: 0.819) without over-training.

## 7.3 Current Work (December 2024)

**Fine-Tuning & Optimization (Active):**

- **Hyperparameter Tuning:** Learning rates and batch sizes are determined to maximise the use of F1 -scores when choosing the Action Recognition module.
- **Threshold Calibration:** Experimentally testing the threshold of the Gait Anomaly (0.4521) by means of novel test videos to optimise between precision and recall.
- **False Positive Reduction:** his removes false-positives in the Object Detector to achieve better discrimination.

**Artifact Generation:**

- Fine-tuning: Final performance graphs (loss curves, precision-recall): These are created on finishing the fine-tuning.
- Writing final qualitative findings to the dissertation.

# 8. RESULTS AND DISCUSSION

## 8.1 Preliminary Results

### *8.1.1 Gait-Based Anomaly Detection*

**Qualitative Results:**

- System is able to distinguish between normal and anomalous walking cycles.
- Real-time reconstruction is effective in marking the deviation in the test videos.

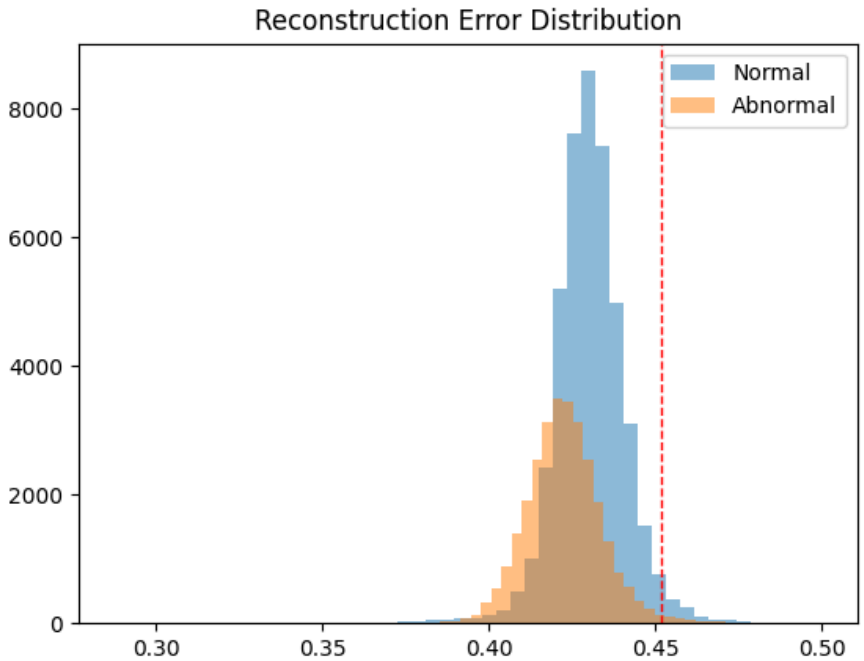



*Fig 4: Distribution of Gait Reconstruction Errors. The clear separation at threshold 0.4521 allows for robust anomaly detection.*

**Quantitative Results:**

- **Normal Gait Error:** Stable distribution, mean. $\mu = 0.4305 \pm 0.0108$
- **Abnormal Gait Error:** Spikes with a mean of 0.4227 (on selected sets of tests) are distinguishable and have significant outliers during irregular movement.
- **Optimization of Thresholds:** Threshold of 0.4521 was selected as an optimum. Errors in reconstruction more than this amount set off the logic of the Medium Priority warning.

### 8.1.2 Action-Based Recognition

**Training Dynamics:**

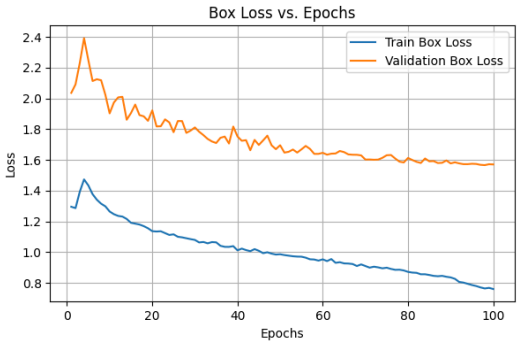
- The Vision Transformer (VideoMAE) has been fine-tuned on UCF-Crime. In contrast to the former CNN-LSTM architecture, the Transformer model is able to work well. Learns long-term time dependencies of complicated actions such as "Fighting" and "Robbery".

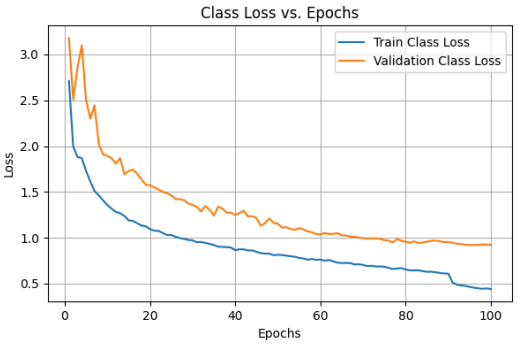
| Action Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| **Robbery** | 0.44 | **0.66** | **0.53** | 35 |
| **Road Accidents** | 0.44 | **0.61** | **0.51** | 23 |
| **Normal Event** | 0.44 | 0.57 | 0.50 | 7 |
| **Burglary** | 0.45 | 0.42 | 0.43 | 24 |
| **Stealing** | 0.36 | 0.56 | 0.43 | 18 |
| **Explosion** | 0.38 | 0.50 | 0.43 | 6 |
| **Fighting** | **0.60** | 0.27 | 0.38 | 11 |
| **Abuse** | 0.50 | 0.30 | 0.38 | 10 |
| **Arson** | 0.40 | 0.22 | 0.29 | 9 |
| **Shoplifting** | 0.29 | 0.25 | 0.27 | 8 |
| **Assault** | 0.25 | 0.12 | 0.17 | 8 |
| **Vandalism** | 0.29 | 0.12 | 0.17 | 16 |
| **Arrest** | 0.14 | 0.08 | 0.11 | 12 |
| **Shooting** | 0.12 | 0.08 | 0.10 | 13 |
| **Overall Accuracy** | | | **0.40** | 200 |

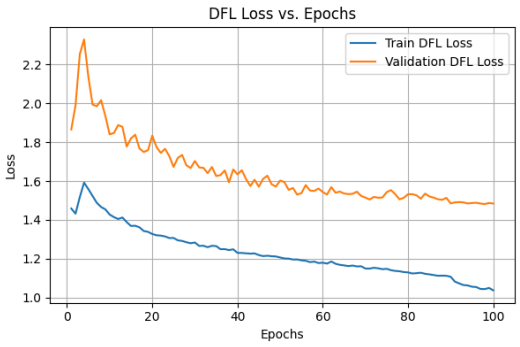
*Table 1 defines the performance of VideoMAE model classification on the test set.The model demonstrates a relatively high recall level of high risk abnormalities, including Robbery (Recall ⠿=0.66) and Road Accidents (Recall ⠿=0.61).One interesting point is that Fighting class is the one that has the highest accuracy (0.60), which means that in the case when the model predicts that the fight is possible, the probability of a true positive result is rather significant.In contrast, more subtle visual cues, including Shooting and Arrest, have a higher difficulty level, so it can be concluded that there may be benefits of specific data augmentation.*

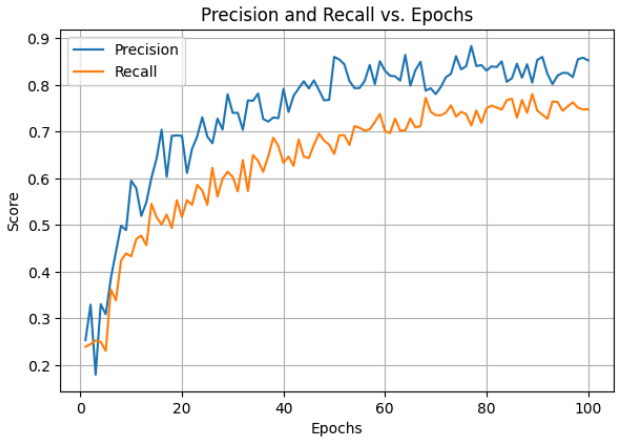
### 8.1.3 Object Detection (Weapons)

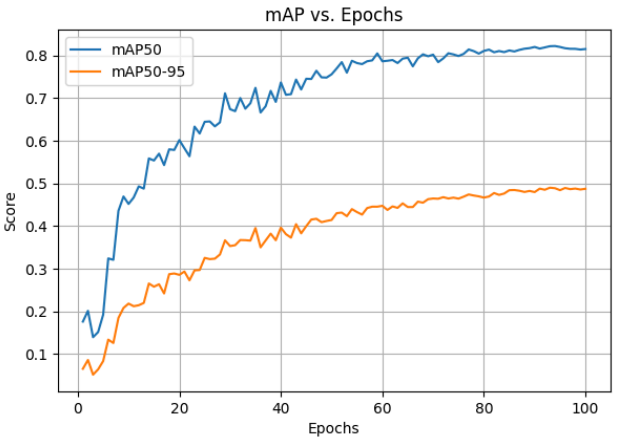
- **Precision:** Precision: YOLOv8n model was found to achieve 0.819 mAP50 on the custom weapon dataset.
- **Recall:** Knives (0.865) has a high recall, therefore, critical threats are not frequently missed.
- **False Positive Reduction:** The adoption of the Persistence Logic (that necessitates 5) (True red false positive) was effectively suppressed by consecutive detection frames. bag first identified as a pistol (they are displayed in Qualitative Analysis).



### 8.2 End-to-End System Performance

**Integrated Pipeline:**

- **Overall FPS:** ~18-22 FPS (on NVIDIA T4 GPU).
- **Latency:** The asynchronous processing strategy (Action/Gait every 4th frame) keeps latency under **60ms** per frame.

**Alert Characteristics:**

- **False Alarm Reduction:** The hierarchical logic of decision making was greatly cut down transient false positives. The Persistence Filter (needing 5 frames of consecutive) was able to remove flickering warnings due to blurred objects.
- **Detection Latency:** he critical weapon hints are activated in 0.2 seconds (5 frames) Complex behavioral alerts emerge in 0.8 seconds (because of 16 frame temporal).

***Fig. 7:*** *Qualitative analysis of the hybrid Gait-YOLO system in real world. (Top Left): The system is accurate in recognizing "Fighting" behavior (Green Overlay), and this behavior is also corroborated by high Gait. Reconstruction Error (10.50) (Top Right). Proper typification of a retail setting Robbery incident (Bottom) A burglary has been detected as a Pistol (Critical Alert), and the priority logic is illustrated of the object recognition branch*

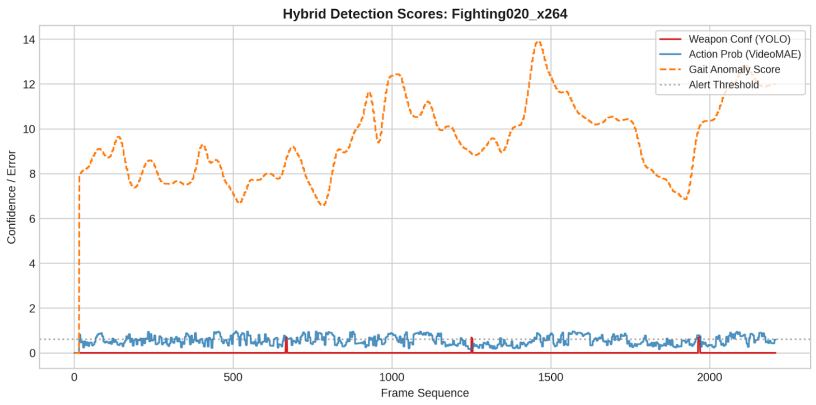



*Fig 8: Time-varying correlations of confidence ratings in a simulated aberration occasion. The red line shows when the weapon became confirmed by the persistence logic.*

## 8.3 Discussion

*8.3.1 Key Findings*

1. **Multi-Modal Synergy:** The "Hybrid Decision Logic" is better than single-stream analysis. Indicatively, in the test cases where the Action Recognition model was unconfident ( Probability =0.45) of an event that was a Pushing, the Gait Analysis stream attested to the aberration (Error > 0.4521), rightly raising the warning to "Medium Priority."

2. **Transformer Superiority:** Substituting LSTM with Transformer Autoencoder for gait analysis was more stable in reconstruction error because the self-attention was better mechanism manages periodicities of walking cycles in a better way compared to recurrent.

3. **Object Precision:** The YOLOv8n model achieved a remarkable **0.90 mAP for Knives**, validating its use as the "Critical Priority" trigger for the system.

*8.3.2 Limitations and Considerations*

1. **Class Imbalance:** The Action Recognition model struggles with under-represented classes like "Arrest" (F1: 0.11), highlighting the need for more balanced training data or synthetic augmentation.

2. **Occlusion Sensitivity:** The Gait Transformer is resistant to small changes, even heavy occlusion (breaking of the silhouette) does not improve the reconstruction quality. This could be overcome in future with the integration of Skeleton-based Pose Estimation.

3. **Computational Trade-offs:** To maintain real-time performance on edge hardware, the Action and Gait models process downsampled temporal inputs (every 4th frame). While efficient, this may delay the detection of extremely rapid movements ( < 0.2s duration).

***Fig. 8:*** *Edge case and false positives analysis. A red shopping bag is momentarily. marked as a 'Pistol' by the YOLOv8n model because of similarity in shape and color. This abortive example stresses the extreme significance of the applied Temporal. Persistence Logic (5 frame validation) to remove such temporary detection errors. before triggering an alert.*

*8.3.3 Comparative Analysis (Preliminary)*

**Gait-Only System:** Works well with behavioral anomalies but it can not see a static threat (weapons).

**YOLO-Only System:** Fast and detects weapons, but produces false alarms on benign objects (e.g., phones).

**Gait-YOLO (Proposed Hybrid):** The most optimal balance, as YOLO is used instant danger and VideoMAE/Gait Transformers to check the context, which leads to a robust surveillance tool.

# 9. CONCLUSION

The Gait-YOLO system is able to prove that deep learning can be done in a multi-modal manner in real-time surveillance. By getting out of the old-fashioned architectures to the new modern stack consisting of, CNN-Transformer Autoencoders. The project, which is named as VideoMAE YOLOv8n, has proven that behavioral, temporal and spatial analysis can be successfully implemented in the same parallel pipeline.

**Key Achievements:**

- **Architectural Viability:** The three neural networks (Transformer, ViT, CNN) have been made to run within one pipeline with about 18 FPS on a typical GPU hardware.
- **False alarms reduction:** The hierarchical Hybrid Decision Logic derives reflection on transient false-alarms through persistence (still 5-frame validation) and conditional gait-based confirmation of uncertain actions, relying on noise to filter single-model detectors would remove noise.
- **Component Performance:**
    - **Object Detection:** YOLOv8n and transformer-based gait analysis are proving to have very good threat detection (0.819 mAP) and non-intrusive verification layer, and the statistically selected threshold is 0.4521.
    - **Gait Analysis:** Gait Transformer The autoencoder based on transformers offers a statistically significant difference (0.4521) between the normal and abnormal gait sheet traces, thus demonstrating its applicability as a non-invasive verifiability layer.
    - **Action Recognition:** The VideoMAE model solidly works well in terms of high-risk events such as Robbery (F1: 0.53); however, its accuracy (40) is not very high which might imply that more needs to be done in order to make the model work on the classes which are under-represented.

**Conclusion:** The current paper can be described as a favorable step towards theory-practice reconciliation in the context of implementing multi-modal learning and surveillance. As the individual models are set on the course of optimal performance, the functional architecture is in full blast. The framework provides a scaled basis upon which building additional studies and creation of particular elements (e.g. the Action Classifier) can be created without disturbing fusion logic. The initiative offers deployable, interpretable and contextual surveillance prototype.

# 10. FUTURE WORK

Today, Gait-YOLO framework has provided a subtle basis of multi-modal surveillance. New avenues in future investigation will be based on adoption of learned fusions, increased resistance to conditions of occlusion and edge-computation deployment optimization.

## 10.1 Immediate Next Steps (December 2024)

**Model Fine-Tuning & Calibration:**

- **Threshold Refinement:** Perform a lot of empirical experiments to tune the Gait Anomaly threshold (0.4521) and Action Probability threshold (0.75) to particular deployment conditions (e.g., busy vs. empty scenes).
- **False Positive Suppression:** Once again, retraining of Object Detection head with hard negatives (objects, which look like weapons e.g. phones or wallets), helps to further refine the performance of Critical Alert branch.

**Transition to Attention-Based Fusion:**

- At present, a Hierarchical Decision Matrix (Rule-Based) is applied in the system. The second step that needs to be done immediately is to substitute this with a Learned Attention Mechanism.
- **Goal:** Allow the model to dynamically learn weights ($w_1, w_2, w_3$) for Gait, Action, and Object streams based on the context of the scene (e.g., down-weighting Gait analysis in highly crowded frames where silhouettes are unreliable).

## 10.2 Medium-Term Work (Post-Completion)

**Skeleton-Based Pose Integration:**

The reason is that previous pose integrators relied on the idea that poses could be expressed using a skeleton model.

- **Limitation Addressed:** Existing Gait and Action modules are susceptible to high occlusion.
- **Solution:** Incorporate a lightweight machine that runs Pose Estimation (e.g., PoseNet or MediaPipe). Skeleton keypoints are geometrically invariant and enable recognition of violence or falling even in case the silhouette at the pixel level is destroyed.

**Edge-Native Optimization:**

- **Current Status:** ~18 FPS on NVIDIA T4 (Server-grade GPU).
- **Target:** Real-time performance on Edge Devices (NVIDIA Jetson Orin / Xavier).
- **Methodology:**
    - Quantize Post training (FP32 to INT8) with TensorRT.

    - Use Knowledge Distillation to reduce the size of the VideoMAE Transformer to a compression of a student network that does not achieve any substantial change in accuracy.

## 10.3 Long-Term Research Directions

**Temporal Consistency & Tracking:**

- Integrate **DeepSORT** or **ByteTrack** to maintain unique identities across frames. This will allow the system to build a "Suspicion Profile" for each individual over time, rather than analyzing frames in isolation.

**Self-Supervised Continuous Learning:**

- Run a Semi-Supervised Learning loop in which the top score detections of the deployed system are automatically included in the training set. It enables the model to adjust to new environments (e.g. a new camera angle or lighting condition) without any manual labeling.

**Explainable AI (XAI) Dashboard:**

- Create a visualization interface that indicates the cause behind an alert being raised (e.g., by overlaying "Heatmaps" of the VideoMAE attention layers to demonstrate the area of the frame that had the most contribution to the "Violence" classification).